



SCHOOL OF COMPUTER TECHNOLOGY

AASD 4008 Big Data Tools and Technologies

What is SSIS

SQL Server Integration Services (SSIS)

SQL Server Integration Services is a component of the Microsoft SQL Server database software that can be used to perform a broad range of data migration tasks.

SSIS is a platform for data integration and workflow applications. It features a data warehousing tool used for data **extraction, transformation, and loading (ETL)**

It's a platform for building enterprise-level data integration and data transformations solutions. Use Integration Services to solve complex business problems by copying or downloading files, loading data warehouses, cleansing and mining data, and managing SQL Server objects and data.

SSIS Includes

1. **a rich set of build-in-tasks and transformations**
2. **Graphical Tools for building packages**
3. **An SSIS Catalog database to store , run and manage packages**



What is SSIS – Continued

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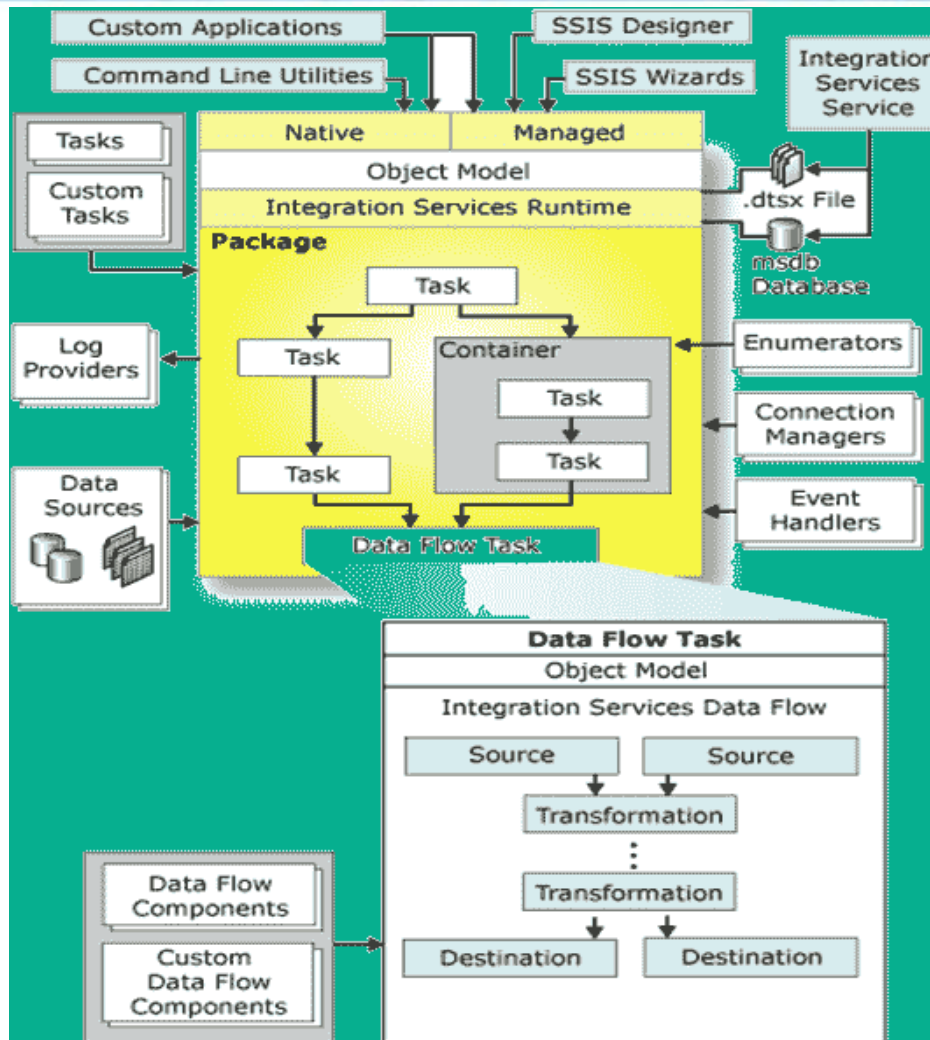
SSIS is mainly used to perform two functionalities

- **Data Integration**

- SSIS performs data integration by combining the data from multiple sources and provides unified data to users

- **Workflow**

- Workflow can be used to perform several things. Sometimes we need to execute some specific steps or a particular path which is either based on the time period or the parameter passed to the package or the data queried from the database.



Components of SSIS

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Control Flow	Control flow is the brain of SSIS package. It helps you to arrange the order of execution of all its components. The components contains <u>Containers</u> and <u>Tasks</u> which are managed by <u>precedence constraint</u>
Precedence Constraints	Precedence constrain are package component which direct tasks to execute in a predefined order. It also defines the workflow of the entire SSIS <u>package</u> . It controls the execution of the two linked tasks by executing the destination tasks based on the result of the earlier task — business rules which are defined using special expressions.
Tasks	A 'Task' is an individual unit of work. It is the same as a method/function used in a programming language. However, in SSIS, you don't use coding methods. Instead, you will use drag & drop technique to design surface and to configure them.
Containers	The container is units for grouping tasks together into units of work. Apart from offering visual consistency, it also allows you to declare variables and event handlers which should be in the scope of that specific container. Following are the types of containers • Sequence container • For Loop • For Each
Data Flow	The main use of this is to Extract data into server's memory, transform it and write it to another destination
Package	It's a collection of tasks which executer in an orderly fashion
Parameters	Parameters behave much like a variable but with a few main exceptions. It can be set outside the package easily. It can be designated as values that must be passed in for the package to start.

Commonly used tasks

Following are some commonly used tasks.

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Execute SQL Tasks	As its name suggests, it will execute a SQL statement against a relational database.
Data Flow tasks	This task can read data from one or more sources. Transform the data when it is in the memory and write it out against one or more destinations. There are further tasks specific to data flow tasks. Explain in next slide
Execute package Task	Use can use this SSIS task to execute other packages from within the same project.
Script Task	You can write NET code which performs any task; you want to perform.
Send Mail Task	You can send an email to notifying users that your package has is finished, or some error occurs.

Commonly used components and transformations – Data Flow



Following are some commonly used tasks.

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Source & Destination	Support various types of source objects (CSV,text, Database...) and various destinations
UNION & Merge	The Union All transformation combines multiple inputs into one output. For example, the outputs from five different Flat File sources can be inputs to the Union All transformation and combined into one output. Union doesn't require data sources to be sorted , merge does Merge can only accept two sources, but union can have multiple
Merge Join	The Merge Join transformation provides an output that is generated by joining two sorted datasets using a FULL, LEFT, or INNER join. For example, you can use a LEFT join to join a table that includes product information with a table that lists the country/region in which a product was manufactured. The result is a table that lists all products and their country/region of origin.
Sort	In Integration Services, the Merge and Merge Join transformations require sorted data for their inputs. The input data must be sorted physically, and sort options must be set on the outputs and the output columns in the source or in the upstream transformation. If the sort options indicate that the data is sorted, but the data is not actually sorted, the results of the merge or merge join operation are unpredictable.
Conditional split	The Conditional Split transformation can route data rows to different outputs depending on the content of the data. The implementation of the Conditional Split transformation is similar to a CASE decision structure in a programming language. The transformation evaluates expressions, and based on the results, directs the data row to the specified output. This transformation also provides a default output, so that if a row matches no expression it is directed to the default output.

Commonly used components and transformations – Data Flow



Following are some commonly used tasks.

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Multicast	The Multicast transformation distributes its input to one or more outputs. This transformation is similar to the Conditional Split transformation. Both transformations direct an input to multiple outputs. The difference between the two is that the Multicast transformation directs every row to every output, and the Conditional Split directs a row to a single output. For more information, see Conditional Split Transformation .
aggregate	The Aggregate transformation applies aggregate functions, such as Average, to column values and copies the results to the transformation output. Besides aggregate functions, the transformation provides the GROUP BY clause, which you can use to specify groups to aggregate across.
Derived Column	The Derived Column transformation creates new column values by applying expressions to transformation input columns. An expression can contain any combination of variables, functions, operators, and columns from the transformation input. The result can be added as a new column or inserted into an existing column as a replacement value. The Derived Column transformation can define multiple derived columns, and any variable or input columns can appear in multiple expressions.
Lookup	The Lookup transformation performs lookups by joining data in input columns with columns in a reference dataset. You use the lookup to access additional information in a related table that is based on values in common columns.
Cache transformation	The Cache Transform transformation generates a reference dataset for the Lookup Transformation by writing data from a connected data source in the data flow to a Cache connection manager. The Lookup transformation performs lookups by joining data in input columns from a connected data source with columns in the reference database.

Demo



Use the following web link and follow the steps to understand basics of SSIS

<https://docs.microsoft.com/en-us/sql/integration-services/lesson-1-create-a-project-and-basic-package-with-ssis?view=sql-server-ver15>

Exercise

<https://www.wiseowl.co.uk/integration-services/exercises/standard/>

<https://www.tutorialgateway.org/ssis/>